

Q-NarwhalKnight Mainnet

Block Production & Reward System

DAG-BFT Consensus with Time-Based Halving

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Abstract

This document provides technical specifications for the Q-NarwhalKnight mainnet, featuring high-throughput DAG-BFT consensus (2-10 blocks/second) and time-based reward halving every 4 calendar years. Initial block reward: 0.05 QUG, halving until reaching 21 million QUG maximum supply over 256 years.

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1 Quick Reference

Parameter	Value
<i>Block Production</i>	
Production Rate	2-10 blocks/second
Consensus Type	DAG-Knight + Narwhal BFT
<i>Rewards</i>	
Block Reward	0.05 QUG
Dev Fee (1%)	0.0005 QUG
Miner Reward (99%)	0.0495 QUG
<i>Halving</i>	
Method	Time-Based (Calendar)
Interval	Every 4 years
First Halving	Oct 26, 2029
<i>Economics</i>	
Maximum Supply	21,000,000 QUG
Genesis Date	Oct 26, 2025 00:00 UTC
Decimals	8 (like Bitcoin)

Table 1: Q-NarwhalKnight mainnet specifications

2 Block Production

2.1 Variable Block Time

Unlike traditional blockchains with fixed block times (e.g., Bitcoin’s 10 minutes), Q-NarwhalKnight produces blocks **continuously** at 2-10 blocks per second:

- **Minimum:** 2 blocks/sec $\rightarrow$ 1 block every 0.5 seconds
- **Average:** 5 blocks/sec $\rightarrow$ 1 block every 0.2 seconds
- **Maximum:** 10 blocks/sec $\rightarrow$ 1 block every 0.1 seconds

2.2 Daily Block Production

Rate	Blocks/Day	Blocks/Year
2 blocks/sec	172,800	63,072,000
5 blocks/sec	432,000	157,680,000
10 blocks/sec	864,000	315,360,000

Table 2: Block production at different throughput levels

3 Block Rewards

Important Note

Q-NarwhalKnight uses **Adaptive Block Rewards** (Section 5.2) as the primary reward system. Fixed rewards were used in early testnet phases but are being phased out in favor of the adaptive system that maintains constant annual emission regardless of network throughput.

3.1 Reward Distribution Structure

Regardless of the reward calculation method (adaptive or fixed), each block distributes rewards as follows:

Recipient	Share	Percentage
Development Fee	1% of block reward	1%
Miner Pool	99% of block reward	99%

Table 3: Reward distribution per block

3.2 Miner Reward Distribution

The miner pool (99% of block reward) is split equally among all mining solutions in the block:

$$\text{Reward per Solution} = \frac{0.99 \times \text{Block Reward}}{\text{Number of Solutions}} \quad (1)$$

Example at 10 blocks/second (with adaptive rewards):

- Block reward: 0.00026 QUG
- Development fee (1%): 0.0000026 QUG
- Miner pool (99%): 0.00025774 QUG
- With 10 solutions: 0.000025774 QUG per solution
- With 100 solutions: 0.0000025774 QUG per solution

3.3 Implementation

See Section 5.2 for the adaptive reward system implementation, which replaces the earlier fixed reward approach. The adaptive system is documented in `crates/q-storage/src/emission_controller.rs`.

4 Time-Based Halving

4.1 Critical Distinction

Time-Based Halving (NOT Block-Based!)

Q-NarwhalKnight uses **calendar time** for halvings, NOT block height!

Why? With variable block production (2-10/sec), block-based halvings would be unpredictable. Time-based halvings provide:

- Predictable economic schedule
- Independent of network throughput
- Clear halving dates
- Resistant to consensus changes

4.2 Halving Formula

From `crates/q-storage/src/balance_consensus.rs`:

```

1 const SECONDS_PER_HALVING: u64 = 126_144_000;
2 // 4 years (365.25 * 4 * 24 * 60 * 60)
3
4 pub const GENESIS_TIMESTAMP: u64 = 1761436800;
5 // Oct 26, 2025 00:00:00 UTC
6
7 let elapsed = current_timestamp - genesis_timestamp;
8 let halving_count = elapsed / SECONDS_PER_HALVING;
9
10 if halving_count >= 64 {
11     return 0; // After 256 years
12 }
13
14 let reward = BASE_REWARD >> halving_count;
```

4.3 Halving Schedule

Era	Date Range	Reward/Block
1	Oct 2025 - Oct 2029	0.0500 QUG
2	Oct 2029 - Oct 2033	0.0250 QUG
3	Oct 2033 - Oct 2037	0.0125 QUG
4	Oct 2037 - Oct 2041	0.00625 QUG
5	Oct 2041 - Oct 2045	0.003125 QUG
...	...	...
64+	After Oct 2281	<0.0001 QUG

Table 4: Time-based halving schedule

5 Emission Analysis

5.1 Why Fixed Rewards Don't Scale

Problem with Fixed Rewards

Early testnet phases used **fixed 0.05 QUG per block**, which creates a fundamental scaling problem:

Fixed Reward $\times$ Variable Throughput = Unpredictable Emission

- At 2 blocks/sec: 3,153,600 QUG/year $\rightarrow$ 21M in 6.7 years
- At 5 blocks/sec: 7,884,000 QUG/year $\rightarrow$ 21M in 2.7 years
- At 10 blocks/sec: 15,768,000 QUG/year $\rightarrow$ 21M in 1.3 years

Solution: Adaptive rewards that scale inversely with throughput!

5.2 Adaptive Block Reward System

Core Innovation: Throughput-Independent Emission

Q-NarwhalKnight uses **adaptive block rewards** to achieve 256-year emission at ANY throughput:

$$\text{Reward per Block} = \frac{\text{Target Annual Emission}}{\text{Blocks Produced This Year}} \quad (2)$$

Result: Reward scales *inversely* with throughput $\rightarrow$ Annual emission stays constant!

- At 10 blocks/sec: 0.00026 QUG/block $\rightarrow$ 82,031 QUG/year
- At 10,000 blocks/sec: 0.00000026 QUG/block $\rightarrow$ 82,031 QUG/year

This enables unlimited throughput scaling (10,000+ blocks/sec) while maintaining predictable 256-year emission schedule.

5.2.1 Layer 1: Time-Based Halving (Predictable Schedule)

Rewards halve every 4 calendar years, regardless of block production rate:

```
1 // From balance_consensus.rs
2 const SECONDS_PER_HALVING: u64 = 126_144_000; // 4 years
3
4 let elapsed_seconds = current_timestamp - GENESIS_TIMESTAMP;
5 let halving_count = elapsed_seconds / SECONDS_PER_HALVING;
6 let reward = BASE_REWARD >> halving_count; // Divide by 2^n
```

Key Properties:

- Halving dates are predictable (Oct 26, 2029, 2033, 2037...)
- Reward stays **constant** during each 4-year period
- Independent of network throughput (2-10 blocks/sec)
- Resistant to consensus upgrades changing block production rate

5.2.2 Layer 2: Supply Cap Enforcement (Hard Limit)

Hard cap prevents total supply from exceeding 21M:

```

1 // From q-types/src/lib.rs
2 pub const QUG_MAX_SUPPLY: u64 = 2_100_000_000_000_000; // 21M
3
4 // From rewards.rs (supply calculation)
5 total_supply.min(self.config.max_supply) // Hard cap at 21M

```

Protection Against High Throughput:

At maximum throughput (10 blocks/sec), network would hit 21M cap in just **486 days**:

$$\text{Time to cap} = \frac{21,000,000 \text{ QUG}}{10 \text{ blocks/sec} \times 0.05 \text{ QUG/block}} = 42,000,000 \text{ sec} = 1.33 \text{ years} \quad (3)$$

5.3 Achieving 256-Year Emission Timeline

For emission to last through 64 halvings (256 years), network must maintain controlled throughput.

Geometric Series Calculation:

$$\begin{aligned}
 \text{Total supply} &= (\text{reward} \times \text{blocks per era}) \times 2 \\
 21,000,000 &= (0.05 \times \text{blocks in 4 years}) \times 2 \\
 \text{blocks in 4 years} &= 210,000,000 \text{ blocks} \\
 \text{Average blocks/sec} &= \frac{210,000,000}{126,144,000} = \mathbf{1.66 \text{ blocks/sec}}
 \end{aligned}$$

Solution: Adaptive Rewards Enable 256-Year Timeline**Key Innovation:** Reward adjusts automatically based on network throughput!**How It Works:**

1. Network measures recent block production rate (trailing 1000 blocks)
2. Calculates expected blocks for the year: $blocks_per_year = rate \times 31,557,600$
3. Adjusts reward: $reward = target_annual_emission \div blocks_per_year$
4. Result: Annual emission stays constant regardless of throughput!

Timeline at Different Throughput (with adaptive rewards):

- 10 blocks/sec: Reward = 0.00026 QUG → 21M in 256 years
- 100 blocks/sec: Reward = 0.000026 QUG → 21M in 256 years
- 10,000 blocks/sec: Reward = 0.00000026 QUG → 21M in 256 years

Benefits:

- Unlimited throughput scaling (no throttling needed!)
- Predictable 256-year emission schedule
- Miner revenue independent of throughput (fair competition)
- Smooth transition to fee-based economy after era 64

5.4 Adaptive Reward Scenarios

Avg Blocks/Sec	Reward/Block	Annual Emission	Time to 21M
1	0.0026 QUG	82,031 QUG	256 years
10	0.00026 QUG	82,031 QUG	256 years
100	0.000026 QUG	82,031 QUG	256 years
1,000	0.0000026 QUG	82,031 QUG	256 years
10,000	0.00000026 QUG	82,031 QUG	256 years

Table 5: Adaptive rewards maintain constant emission across all throughputs

Key Insight: With adaptive rewards, annual emission is *invariant* to throughput. Network can scale to any performance level without affecting the 256-year emission schedule!

5.5 Fee Market Integration

Critical Economic Shift at High Throughput

As network scales to high throughput, **transaction fees become the primary miner incentive**. Block rewards decrease proportionally with throughput.

At 10,000 blocks/sec, reward per block is only 0.00000026 QUG. Miners depend on fees for profitability:

Key Insight: Annual emission stays constant (82,031 QUG/year) regardless of throughput. At high throughput, miners compete for transaction fees rather than block rewards.

Throughput	Reward/Block	Revenue (1% share)	Fee Priority
10 bps	0.00026 QUG	2.2 QUG/day	Minimal
100 bps	0.000026 QUG	2.2 QUG/day	Low
1,000 bps	0.0000026 QUG	2.2 QUG/day	Medium
10,000 bps	0.00000026 QUG	2.2 QUG/day	Primary

Table 6: Fee market economics across throughput levels

Dynamic Fee Adjustment: Minimum transaction fee scales automatically:

$$\text{min_fee} = \text{BASE_FEE} \times \sqrt{\frac{\text{BASE_REWARD}}{\text{adaptive_reward}}} \quad (4)$$

This ensures transaction fees remain economically viable for miners as block rewards decrease with throughput.

5.6 Migration Strategy

Phased Activation for Network Stability

The adaptive reward system activates at **block height 200,000** (~90 days after mainnet launch):

Phase 1 (Blocks 0-199,999): Bootstrap Period

- Fixed reward: 0.05 QUG/block (legacy Phase 8 economics)
- Allows miners to update software
- Ensures network stability during initial launch
- EmissionController tracks throughput but doesn't control rewards yet

Phase 2 (Block 200,000+): Adaptive Rewards

- Pure adaptive: Reward $\propto 1/\text{throughput}$
- Annual emission locked to 82,031 QUG/year (Era 1)
- Nodes running old software will reject blocks (forced upgrade)
- Network can scale to 10,000+ blocks/sec without economic impact

Grace Period Benefits:

- 90-day window for miners to upgrade software
- Smooth transition reduces network disruption
- Backward compatibility maintained during bootstrap
- Clear activation block allows coordinated upgrade

Implementation: The EmissionController (Section 10, Table 11) implements dual-phase emission with automatic transition based on block height.

Blocks/Day	QUG/Day	QUG/Month	QUG/Year
10	0.495	14.85	180.68
50	2.475	74.25	903.38
100	4.950	148.50	1,806.75
500	24.750	742.50	9,033.75

Table 7: Solo mining profitability (0.0495 QUG per block after dev fee)

6 Mining Profitability

6.1 Solo Mining

If you mine complete blocks (all solutions yours):

6.2 Pool Mining

Network share percentage of total daily emission:

Share	QUG/Day	QUG/Month	QUG/Year
1%	216	6,480	78,840
5%	1,080	32,400	394,200
10%	2,160	64,800	788,400
25%	5,400	162,000	1,971,000

Table 8: Pool mining at 5 blocks/sec (21,600 QUG/day total)

7 Why Block-Based Halving Fails with DAG-BFT

Problem: Block-Based Halving with High Throughput

Bitcoin's halving occurs every 210,000 blocks (~4 years at 10-minute block time). With DAG-BFT producing 2-10 blocks/second, block-based halving would create economic chaos:

Calculation: $210,000 \text{ blocks} \div 10 \text{ blocks/sec} \div 3600 = 5.8 \text{ hours}$

Halvings every 6 hours would make economic planning impossible!

7.1 Time-Based Solution

Q-NarwhalKnight solves this by making halving dependent on **calendar time** instead of block height:

Property	Block-Based	Time-Based
Predictability	Variable (depends on speed)	Fixed calendar dates
Planning	Difficult	Easy
Network upgrades	Affect schedule	No impact
Throughput scaling	Accelerates halvings	No effect
Economic stability	Low	High

Table 9: Block-based vs time-based halving comparison

Example: With DAG-BFT improvements doubling throughput:

- **Block-based:** Next halving in 6 months instead of 1 year (unpredictable)
- **Time-based:** Next halving still on Oct 26, 2029 (predictable)

8 Comparison with Bitcoin

Property	Bitcoin	Q-NarwhalKnight
Block Time	10 minutes	0.1-0.5 seconds
Halving Interval	Block-based 210k blocks	Time-based 4 years
Initial Reward	50 BTC	0.05 QUG
Max Supply	21M BTC	21M QUG
Decimals	8	8
Consensus	PoW	DAG-BFT + VDF
Finality	Probabilistic	Deterministic
Throughput	7 TPS	1000+ TPS

Table 10: Economic comparison

9 Key Takeaways

1. **Variable Block Production:** 2-10 blocks/second (DAG-BFT consensus)
2. **Fixed Block Reward:** 0.05 QUG per block (1% dev fee, 99% miners)
3. **Time-Based Halving:** Every 4 calendar years (independent of block count)
4. **Two-Layer Protection:**
 - Time-based halving controls *when* rewards decrease
 - Supply cap enforcement controls *total* emission (21M hard limit)
5. **Genesis:** October 26, 2025 00:00 UTC
6. **Target Throughput:** ~ 1.66 blocks/sec average for 256-year emission
7. **Current vs Target:** 2-10 blocks/sec requires adaptive throttling
8. **Economic Model:** Predictable halving schedule with asymptotic 21M supply

Core Innovation

Adaptive block rewards enable throughput-independent emission, achieving:

- **Time-based halving:** Predictable emission schedule (Oct 26, 2029, 2033, 2037...)
- **Adaptive rewards:** Automatic adjustment based on network throughput
- **Unlimited scaling:** Network can reach 10,000+ blocks/sec without affecting time-line
- **Fair economics:** Miner revenue independent of throughput (competition-based)

Mathematical Guarantee: Formula $Reward = Annual_Target \div Blocks_Per_Year$ ensures 21M QUG is reached in exactly 256 years (64 halvings) at ANY throughput level. This decouples network performance from monetary policy.

10 Source Code References

Parameter	File	Lines
Adaptive Reward Controller	q-storage/src/emission_controller.rs	1-468
Block Reward Calculation	q-storage/src/balance_consensus.rs	562-576
Time-Based Halving	q-storage/src/emission_controller.rs	205-230
Genesis Time	q-storage/src/emission_controller.rs	19
Dev Fee	q-api-server/src/block_producer.rs	387
Max Supply Constant	q-types/src/lib.rs	91
Supply Cap Enforcement	q-storage/src/emission_controller.rs	248-252

Table 11: Source code locations for adaptive emission control system

10.1 Code Verification

All values in this document are extracted directly from the Q-NarwhalKnight source code. The adaptive reward system is the **primary implementation** (v0.9.99-beta+):

```

1 // Adaptive block reward (q-storage/src/emission_controller.rs)
2 pub const BASE_ANNUAL_EMISSION: u64 = 82_031_000_000_000;
3 // 82,031 QUG/year target (Era 1)
4
5 pub fn calculate_block_reward(
6     &mut self,
7     current_timestamp: u64,
8     total_supply: u64,
9 ) -> Result<u64> {
10     let block_rate = self.calculate_economic_rate();
11     let reward = (annual_target * PRECISION)
12                 / (block_rate * SECONDS_PER_YEAR) as u128;
13     Ok(reward as u64)
14 }
15
16 // Time-based halving interval
17 const SECONDS_PER_HALVING: u64 = 126_144_000; // 4 years
18
19 // Supply cap (q-types/src/lib.rs)
20 pub const QUG_MAX_SUPPLY: u64 = 2_100_000_000_000_000; // 21M

```

Q-NarwhalKnight Mainnet

Time-based halving | 0.05 QUG initial reward | 21M max supply
Genesis: Oct 26, 2025 00:00 UTC

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